

DESCRIPTIVE STATISTICS :

Data की CENTRE बताता है।

I. MEASURE OF CENTRAL TENDENCY

It gives the representative value around which the entire dataset tends to cluster.

1. MEAN

2. MEDIAN

3. MODE

1. MEAN (OR AVERAGE) :-

$$\bar{X} = \left(\frac{\text{Sum of Observations}}{\text{Total no. of observations}} \right)$$

Eq. :- OBSERVATIONS, $X = \{2, 4, 6, 8, 10\}$

$$\bar{X} = \left(\frac{2+4+6+8+10}{5} \right) = 6$$

∴ We calculate mean for NUMERICAL VARIABLES.

* LIMITATION :-

Affected by OUTLIERS or EXTREME VALUES.

Eq. :- $\{2, 4, 6, 8, 10, 100\}$, now mean, $\bar{X} = 21.67$

OUTLIER affecting previous mean BADLY.

2. MEDIAN

- Middle value of the dataset when it is ordered.
- Applicable $\rightarrow$ NUMERICAL VARIABLES.
- Less influenced by OUTLIERS.

MEDIAN $\rightarrow$

- $\rightarrow n$ is ODD:
 - $\text{MEDIAN} = \left(\frac{n+1}{2}\right)^{\text{th}}$ observation
- $\rightarrow n$ is EVEN:
 - $\text{MEDIAN} = \left\{ \frac{\left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n}{2} + 1\right)^{\text{th}}}{2} \right\}$ observation

or

* The average of the middle two values.

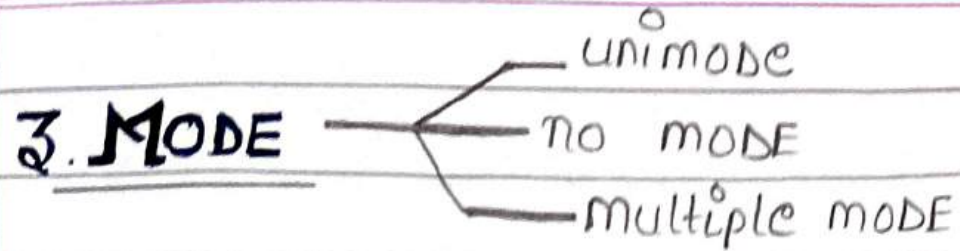
E.g. :- $x = \{3, 1, 7, 5, 9\}$
 arrange $\rightarrow \{1, 3, 5, 7, 9\} \Rightarrow 5 \text{ terms} \Rightarrow \text{ODD}$

$$\therefore \text{median} = 3^{\text{rd}} \text{ obseru}^n = (5)$$

E.g: $x = \{1, 3, 5, 7, 9, 11\} \Rightarrow 6 \text{ terms} \Rightarrow \text{EVEN}$

$$\therefore \text{median} = \left(\frac{5+1}{2}\right) = \left(\frac{12}{2}\right) = (6)$$

Date:



SURVEY → FAVOURITE FRUIT among
APPLE, BANANA, ORANGES :-

∴ MODE = FRUIT mentioned ↑ frequently

- MOST FREQUENT value.
- Applicable BEST FOR CATEGORICAL VARIABLES.
- In case of NUMERICAL VARIABLES → Unique Count should be less.

E.g. ∴ $X = \{2, 4, 3, 4, 2, 6, 7, 2\}$

⇒ (2) ≡ MODE.